Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

March 3, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and NPF-96
NRC Docket No. 50-390 and 50-391

Subject: **Licensee Event Report 390/2017-003-00, Inadequate Operability Determination Leads to a Condition Prohibited by the Technical Specifications**

This submittal provides Licensee Event Report (LER) 390/2017-003-00. This LER provides details concerning an error made while performing an operability evaluation that led to a condition prohibited by the Technical Specifications. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B).

There are no regulatory commitments contained in this letter. Please direct any questions concerning this matter to Gordon Arent, WBN Licensing Director, at (423) 365-2004.

Respectfully,

Paul Simmons
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant

NRC FORM 366 (06-2016) | U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Watts Bar Nuclear Plant, Unit 1	05000390	1 of 5

4. TITLE

Inadequate Operability Determination Leads to a Condition Prohibited by the Technical Specifications

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
									Watts Bar Nuclear Plant, Unit 2	05000391
01	04	2017	2017	- 003	- 00	03	03	2017	FACILITY NAME	DOCKET NUMBER

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Dean Baker, Licensing Engineer	423-452-4589

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BI	FCV	EPG	Y					

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO

15. EXPECTED SUBMISSION DATE MONTH | DAY | YEAR

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On January 4, 2017 at 1010 Eastern Standard Time (EST), Watts Bar Nuclear Plant Operations personnel declared Essential Raw Cooling Water (ERCW) strainer flush valve 2-FCV-67-9B inoperable due to having a through-wall leak. The valve was replaced and the ERCW strainer was returned to service on January 5, 2017 at 0952 EST. This event is reportable because the valve had had a through-wall leak since January 31, 2016 and had not been declared inoperable. With a through-wall leak, a flaw evaluation is required to be performed to demonstrate the through-wall leak was stable. The failure to perform an adequate operability evaluation allowed the valve to remain in service for a period of time longer than allowed by Technical Specification (TS) 3.7.8, Essential Raw Cooling Water, Limiting Condition for Operation (LCO) Condition A. This represents a condition prohibited by the TS. Subsequent analysis of the valve demonstrated that it remained structurally sound with the leak, and would not have impacted the operability of the ERCW system.

The cause of the failure to perform an adequate operability evaluation has been determined to be human performance errors on the part of both operations and engineering personnel. Additional training of operations and engineering personnel is planned as corrective action to address the potential for recurrence of this issue.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000390	2017	- 003	- 00

NARRATIVE

I. PLANT OPERATING CONDITIONS BEFORE THE EVENT

Watts Bar Nuclear Plant (WBN) Units 1 and 2 were at 100 percent rated thermal power (RTP) .

II. DESCRIPTION OF EVENT

A. Event Summary

On January 4, 2017 at 1010 Eastern Standard Time (EST), Watts Bar Nuclear Plant Operations personnel declared Essential Raw Cooling Water (ERCW) {EIIS:BI} strainer flush valve 2-FCV-67-9B {EIIS:FCV} inoperable due to having a through-wall leak. The valve was replaced and the ERCW was returned to service on January 5, 2017 at 0952 EST. This event is reportable because the valve had had a through-wall leak since January 31, 2016 and had not been declared inoperable. A flaw evaluation is required in this situation to demonstrate the through-wall leak was stable and would not become worse. The failure to perform an adequate operability evaluation allowed the valve to remain in service for a period of time longer than allowed by Technical Specification (TS) 3.7.8, Essential Raw Cooling Water, Limiting Condition for Operation (LCO), Condition A.

This event is being reported to the Nuclear Regulatory Commission (NRC) under 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications.

B. Inoperable Structures, Components, or Systems that Contributed to the Event

No additional inoperable systems or components beyond the one identified valve contributed to this report.

C. Dates and Approximate Times of Occurrences

Date	Time (EST)	Event
1/31/2016		Condition Report (CR) 1131468 documents a through-wall leak in the body of 2-FCV-67-9B.
1/04/2017	1010	Technical Requirement 3.4.5 Piping System Structural Integrity determined not met for 2-FCV-67-9B. ERCW TS LCO 3.7.8 Condition A entered.
1/05/2017	0930	Work Order 117564121 completed to replace valve 2-FCV-67-9B.
1/05/2017	0952	ERCW TS LCO 3.7.8 Condition A exited.

D. Manufacturer and Model Number of Components that Failed During the Event

The leaking valve was a four inch 150 pound class carbon steel ball valve provided by Energy Products Group with a rated operating pressure of 200 psig.

NARRATIVE

E. Other Systems or Secondary Functions Affected

No other systems or secondary functions were affected .

F. Method of discovery of each Component or System Failure or Procedural Error

As a result of an NRC question, the operability of valve 2-FCV-67-9B was reviewed by plant management. The valve was subsequently declared inoperable and the appropriate Technical Requirements (TR) and TS LCO Conditions were entered.

G. Failure Mode and Effect of Each Failed Component

While a through-wall leak was identified for 2-FCV-67-9B, subsequent analysis performed demonstrated that it maintained operability with the presence of a leak.

H. Operator Actions

Upon discovery, Operations personnel promptly entered the appropriate TR and TS LCO conditions.

I. Automatically and Manually Initiated Safety System Responses

There were no safety system responses associated with this issue.

III. CAUSE OF THE EVENT

A. The cause of each component or system failure or personnel error, if known.

The most probable cause of the thru wall leak on valve 2-FCV-67-9B is multimode degradation due to cavitation and jet impingement.

B. The cause(s) and circumstances for each human performance related root cause.

The causes of the inadequate operability evaluation are associated with human performance errors by both operations and engineering personnel. The apparent cause was attributed to a knowledge gap regarding ASME Class 2 and 3 piping associated with TR 3.4.5.

IV. ANALYSIS OF THE EVENT

On January 4, 2017, as a result of an NRC question, the operability of 2-FCV-67-9B was reviewed by plant management. The valve was subsequently declared inoperable and the correct TR and TS entered for the condition. The valve was replaced within 24 hours and the system restored to operability.

In investigating this issue, it was determined that a through-wall leak had been identified by plant personnel on January 31, 2016, and Condition Report (CR) 1131468 was initiated. WBN TR 3.4.5, Piping System Structural Integrity, requires the structural integrity of American Society of Mechanical Engineers (ASME) Code Class 1, 2, and 3 piping at all times. The ERCW system is ASME Code Class 3 piping. The

NARRATIVE

NRC further specifies requirements for through-wall leakage in Inspection Manual Chapter (IMC) 0326, Operability Determinations & Functionality Assessments for Conditions Adverse to Quality of Safety. IMC 0326 indicates for through-wall leakage of Class 3 piping and components, a flaw evaluation would need to be performed. When the leak was identified, the Senior Reactor Operator performing the operability review did not recognize the need to apply TR 3.4.5 and did not request engineering to perform a prompt operability evaluation.

In addition, during the screening process for CR 1131468, site engineering personnel also failed to recognize that a pressure boundary leak of Class 3 components required an ASME Code evaluation.

V. ASSESSMENT OF SAFETY CONSEQUENCES

The valve with the through-wall leak was analyzed after its replacement. This analysis determined that it remained structurally sound, and could accommodate all required design loads, including a safe shutdown earthquake. Accordingly, the safety consequences of this event are low.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The leaking valve was evaluated after removal. This evaluation determined that it had adequate structural integrity to support operability even with a through-wall leak.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not applicable.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

The valve with the through-wall leak was first identified on January 31, 2016 and was not replaced until January 5, 2017. An analysis on the valve after it was removed demonstrates that it would not have impacted operability of the ERCW system.

VI. CORRECTIVE ACTIONS

This event was entered into the Tennessee Valley Authority (TVA) Corrective Action Program and is being tracked under CR 1247701.

A. Immediate Corrective Actions

When the condition was identified, the appropriate TR and TS LCO Conditions were entered. The valve was replaced and the ERCW strainer returned to service. The replacement valve has a stainless steel body, reducing the potential to be impacted by the identified degradation mechanisms.

NARRATIVE

B. Corrective Actions to Prevent Recurrence or to Reduce Probability of Similar Events Occurring in the Future

Additional training of operations and engineering personnel will be conducted related to this issue. In addition, a training program on TS usage and compliance is in the process of being implemented for all licensed operating staff.

VII. PREVIOUS SIMILAR EVENTS AT THE SAME SITE

LER 390/2016-009 reported a condition where TS 3.6.3 for an inoperable containment isolation valve (CIV) was not entered when a surveillance associated with impacted penetration was not performed. The surveillance was recognized as potentially going late and corrective action was not taken in a timely fashion.

LER 390/2016-002 reported a condition where the Note associated with TS 3.6.3 for CIVs was misinterpreted. When a CIV was found inoperable, it was not isolated in four hours as required, rather operations personnel were stationed to manually isolate the penetration if required. This was a misinterpretation of the wording of the note, which allows for intermittent opening under administrative controls of an inoperable penetration to perform testing.

Both this event and these previous events document knowledge gaps related to the TS compliance by members of the WBN operations staff. Specific training is in progress to address these knowledge gaps to reduce the potential for future recurrence.

VIII. ADDITIONAL INFORMATION

None.

IX. COMMITMENTS

None.